The `concept-factors` branch

From CoSpRo to Spatial Concept Distillation:
architectures, numbers and lessons of 26 September to 1 November 2026

Report for the supervisor meeting of 10 November 2026

Abstract

This report explains what the `concept-factors` branch built after the CoSpRo paper, why each design looks the way it does and what each experiment showed. The branch replaced CoSpRo's prior about *which* concept is spurious by a symmetric goal: make the self-supervised student encode every concept CLIP/SpLiCE finds, so that a group-balanced linear probe can pick the side the task needs. Nine architectures were tried in three phases. Image-level concept distillation (F2, F3 and the cross-fitted ridge "xfit") doubled worst-group accuracy (WGA) on Spur-CIFAR10 and taught us that a small dataset lets the student memorize any per-image target. Spatial Concept Distillation (SCD) supervises every location of the ResNet's 7×7 feature map with dense CLIP concept maps; it raises validation WGA over SimCLR by 6.1 (Waterbirds), 6.8 (MetaShift) and 22.5 points (Spur-CIFAR10) and beats its shuffled-map control on all three datasets. Four extensions on top of SCD tried to make the student separate an object from its usual context. Only object compositing, which uses a text prior that types concepts as objects or scenes, adds a large gain (Waterbirds 59.1 WGA); the prior-free variants (region contrast, symmetric mixing and per-group heads) match or fall below SCD. All numbers are validation numbers; locked test runs are prepared and outstanding.

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1 The result on one page

Table 1: Headline validation numbers (mean of the last four linear probes, epochs 425–500; ± is the standard deviation over student seeds; seeds in brackets). "Prior-free" means that no step of the method decides which concept is the class and which the spurious attribute.

	Waterbirds	MetaShift	Spur-CIFAR10	CelebA
SimCLR (reference)	46.7 ± 1.9 (6)	44.3 ± 7.3 (6)	15.6 ± 1.0 (2)	84.9 ± 0.0 (2)
CoSpRo (September paper)	50.0 ± 3.2 (4)	42.9 ± 2.7 (10)	19.6 (1)	–
SCD, prior-free (this branch)	52.8 ± 4.7 (6)	51.1 ± 5.4 (6)	38.1 ± 0.5 (2)	no maps yet
same loss, shuffled maps	47.5 ± 2.5 (6)	45.1 ± 2.9 (6)	17.2 ± 2.0 (2)	
SCD + compositing, object prior	59.1 ± 7.1 (6)	52.3 ± 1.9 (4)	–	excluded

Three sentences to say first

1. CoSpRo helped on Waterbirds only and carried a hidden assumption (the spurious attribute is a removable context concept); the new branch drops that assumption and distils *all* concepts.
2. Distilling concepts per image fails on small datasets because the student memorizes per-image targets; distilling them *per location* of the feature map (SCD) works on all three datasets and beats a control that uses the same loss on another image's maps.
3. SCD makes the core feature stronger, but the student still ties the object to its usual context; the only extension that breaks this tie (compositing) needs an object-versus-scene prior.

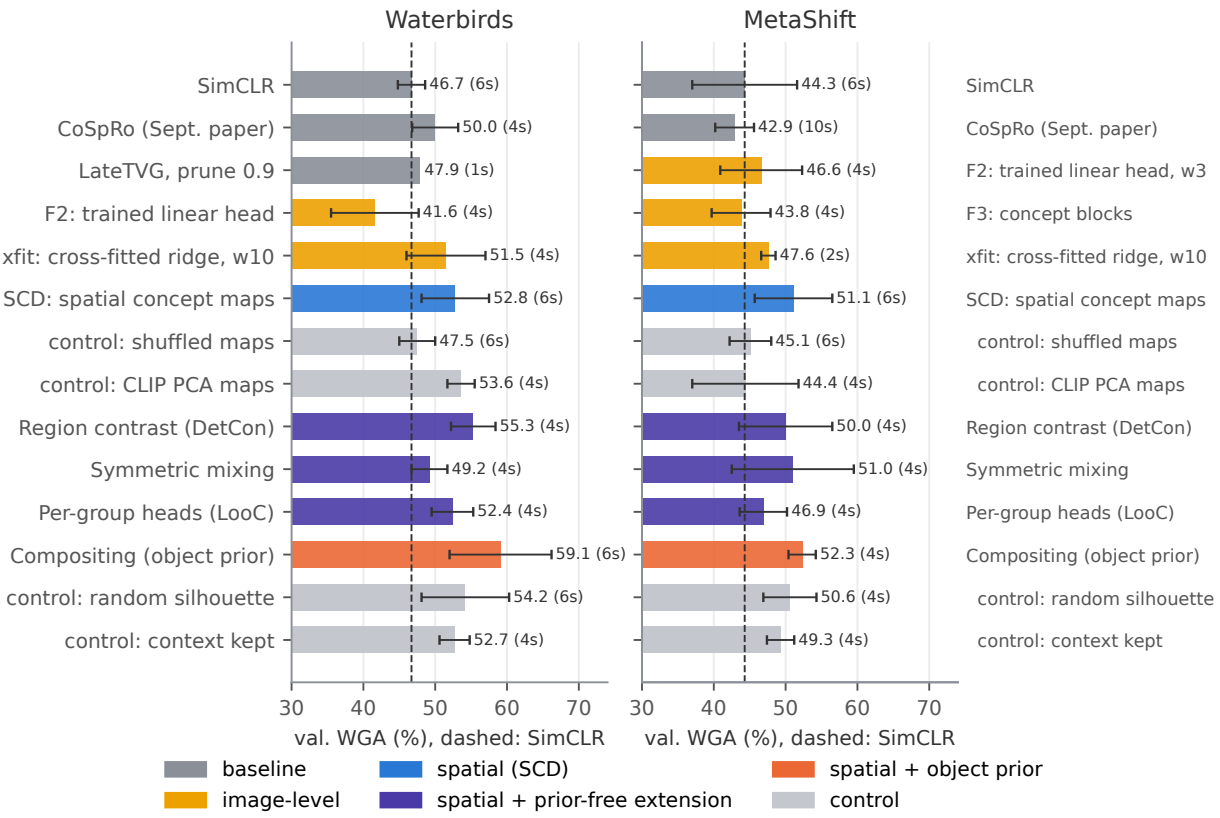


Figure 1: Every architecture of the branch on the two main datasets. Bars are validation WGA, whiskers one standard deviation over seeds, "(6s)" the number of seeds. Grey bars are controls. The dashed line is SimCLR. Section 9 discusses this figure row by row.

2 The problem and what we want to beat

2.1 Spurious correlations and worst-group accuracy

In Waterbirds the bird type agrees with the background on 95% of the training images (waterbirds on water, landbirds on land). In MetaShift 88% of the training cats sit indoors and 88% of the dogs outdoors. A representation that learns "water" as evidence for "waterbird" or "indoor" as evidence for "cat" fails on the rare images that break the rule. Each image belongs to a group $g = (y, a)$ of class y and spurious attribute a and the metric is the accuracy of the worst group:

$$\mathrm{WGA} = \min_g \frac{1}{n_g} \sum_{i:\, g_i = g} \mathbf{1}\{\hat{y}_i = y_i\}. \tag{1}$$

Table 2: Datasets. MetaShift and Spur-CIFAR10 were the development datasets; Waterbirds and CelebA received frozen configurations.

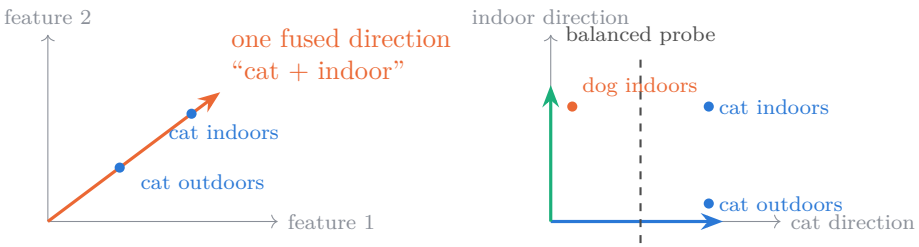
Dataset	Train images	Spurious structure and evaluation
Waterbirds	4,795	bird type vs. land/water background; validation groups 467 / 466 / 133 / 133 images, the two waterbird groups smallest; test 642 per minority group
MetaShift (cats/dogs)	1,700	cat/dog vs. indoor/outdoor, 88% agreement; 72 validation images per group, so WGA is noisy (one image = 1.4 points)
Spur-CIFAR10	45,000	10 CIFAR classes, a 2-pixel coloured line agrees with the class on 95% of the training images; 100 (class, colour) groups
CelebA	162,770	blond hair vs. gender; only image-level runs so far

2.2 Why a linear probe on a balanced set is enough: deep feature reweighting

Our evaluation fits a logistic regression on frozen features using a small *group-balanced* subset of the training split (224 images on Waterbirds, 400 on MetaShift, 1,400 on Spur-CIFAR10). Kirichenko et al. (DFR, ICLR 2023) showed that such a balanced last layer recovers high WGA *whenever the representation encodes the core feature separately from the spurious one*. This fixes the goal of the whole branch:

Goal derived from DFR

The representation must contain a **bird direction** and a **background direction** as two separate, linearly readable directions. It does not have to forget the background. The balanced probe then puts its weight on the bird direction. The failure we must prevent is the *fusion* of the two into one direction ("waterbird-and-water"), or the *suppression* of the harder feature by the easier one.



fused: the probe cannot pick **separate: the probe splits on "cat"**

Figure 2: The geometry behind the goal. Left: a fused direction places cats outdoors close to the origin, so no linear probe separates them from dogs indoors. Right: with separate directions the balanced probe uses the cat axis alone.

2.3 The restriction: no prior about which concept is spurious

Label-free methods against spurious correlations need some assumption about what is spurious. LateTVG and LA-SSL assume the spurious feature is the *easy* one (learned early or by early layers). Background augmentation assumes it is the *background*. CoSpRo assumed it is a *removable context concept*. These assumptions differ between datasets: in CelebA the spurious attribute (gender) is not a background at all. The branch therefore adopts one restriction:

Restriction of the thesis method

No step of pretraining may use target labels, group labels or knowledge of which concept is the class and which the spurious attribute. Every concept enters the loss in the same way. Labels enter only the balanced probe and the post-hoc analysis, identically for every method.

2.4 What we compare against

- **SimCLR** under the identical protocol: the reference every gain is measured against.
- **A control for every new loss**: the same loss on another image's targets ("shuffled"). It keeps every statistic of the targets and removes the link to the image content, so a gain above it is a gain of the concept content. Most decisions of the branch rest on these controls.
- **CoSpRo** (our September paper): 50.64 ± 1.97 test WGA on Waterbirds against 46.82 ± 3.65 for SimCLR; on MetaShift 42.9 ± 2.7 against 44.6 ± 6.9 over ten seeds.
- **LateTVG** (Hamidieh et al., ICLR 2024) reports 55.4 (Waterbirds) and 59.3 (MetaShift) WGA with SimCLR under its own protocol. Under our protocol one seed reaches 47.9 on Waterbirds; the matched six-seed test run is predeclared (`waterbirds_latetvg_test.yaml`).
- **LA-SSL** (learning-speed-aware sampling) has an adaptation in the CoSpRo paper's appendix.

Absolute accuracies are low (SimCLR reaches 52% average accuracy on Waterbirds) because a ResNet-18 trains from random initialization on 1,700 to 4,795 images. Supervised or ImageNet-pretrained numbers (DFR reaches above 90% WGA on Waterbirds with a pretrained ResNet-50) answer a different question: our protocol measures what the SSL objective learns from the dataset's own images.

3 Where we started: CoSpRo in two paragraphs

CoSpRo removed the subspace of one concept group from the CLIP embeddings, kept image pairs that became closer, filtered them (activation contrast, residual concept agreement, synthetic null groups) into a sparse teacher graph and transferred the graph to a SimCLR student through graph-aware batches and a relational KL loss. On Waterbirds it gained about four points of test WGA.

Two observations ended this line. First, the gain did not transfer: below SimCLR on MetaShift over ten seeds, 19.6 against 15.6 on one Spur-CIFAR10 seed. Second, the method carries the prior that the spurious attribute is a removable context concept, the graph keeps wrong-class edges and the background stays linearly decodable at 91%.

4 Shared machinery of every architecture

4.1 Training and evaluation protocol

Table 3: Protocol shared by every run of the branch (inherited from CoSpRo so that numbers are comparable).

Student	ResNet-18 from random initialization. Waterbirds, MetaShift, CelebA: 224-pixel input ("large" stem, 7×7 convolution with stride 2 and max-pooling). Spur-CIFAR10: 32-pixel input (CIFAR stem)
SSL	SimCLR, 500 epochs (CelebA 250), batch 128, temperature 0.05, SGD with learning rate 0.01 decayed by 0.1 at epochs 350, 400 and 450, weight decay 10^{-4}, crop scale $[0.2, 1]$, float16 autocast with a GradScaler on one V100
Probe	logistic regression (L2 coefficient 0.001) on the pooled 512-d features, fitted on the group-balanced subset of the training split, evaluated every 25 epochs
Reported value	mean of the last four probes (epochs 425 to 500); this damps single-probe noise on MetaShift's 72-image groups
Seeds	4 to 6 for Waterbirds and MetaShift, 2 for Spur-CIFAR10 and CelebA
Split	validation; the locked test split stays unused until the configuration is final

4.2 The ResNet-18 student and why we read the 7×7 map

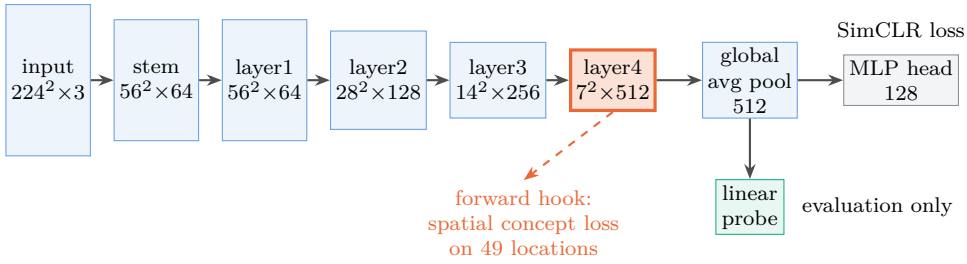


Figure 3: The student. SimCLR and the probe read the 512-d vector after global average pooling. The spatial concept loss reads the $7 \times 7 \times 512$ output of `layer4` through a forward hook, *before* pooling. Spur-CIFAR10 uses the CIFAR stem, so its map is $4 \times 4 \times 512$.

Why the 7×7 map before pooling and not the pooled vector

- **Pooling is where concepts fuse.** After global average pooling a cat and a couch become one 512-d vector per image. Across images they co-occur (88%), so an image-level target cannot tell "cat" from "couch" except on the few images where they disagree. At the 7×7 map they sit at different locations of *every* image.
- **The last stage is what the probe reads.** `layer4` feeds the pooled vector directly, so shaping it shapes exactly the representation the probe evaluates. An earlier stage would be shaped and then transformed by `layer4` before the probe sees it.
- $7 = 224/32$. ResNet-18 halves the resolution five times (stem convolution, max-pooling, `layer2`, `layer3`, `layer4`). Nothing is cropped or "thrown away"; we tap the tensor that pooling would average.
- **Limitation built into this choice.** Each of the 49 locations has a receptive field that covers most of the 224-pixel view. A "cat" location therefore also sees the scene, which is why the student keeps part of the co-occurrence (Section 7.6).

4.3 SpLiCE concepts

A frozen OpenCLIP ViT-B/32 (LAION-2B weights) embeds every training image as z_i. SpLiCE (Bhalla et al., NeurIPS 2024) writes the centred, normalized embedding as a sparse non-negative combination of text directions from a vocabulary,

$$u_i = \mathrm{norm}(z_i - \mu), \qquad c_i \in \arg\min_{c \geq 0} \|u_i - D^\top c\|_2^2 + \lambda_1 \|c\|_1, \tag{2}$$

with L1 penalty 0.25. About nine concepts are active per image. MetaShift uses the LAION vocabulary (10,000 words), Waterbirds and Spur-CIFAR10 the Open Images v7 vocabulary (20,931 words).

4.4 From concepts to concept factors: the meaning grouping

A vocabulary names one visual content several times (*cat, kitten, kitty*). A *concept factor* is a group of concepts that name one thing; its value on an image is the sum of its members' SpLiCE codes. Getting this grouping right took a full round of the branch, because the obvious signals fail (Figure 4).

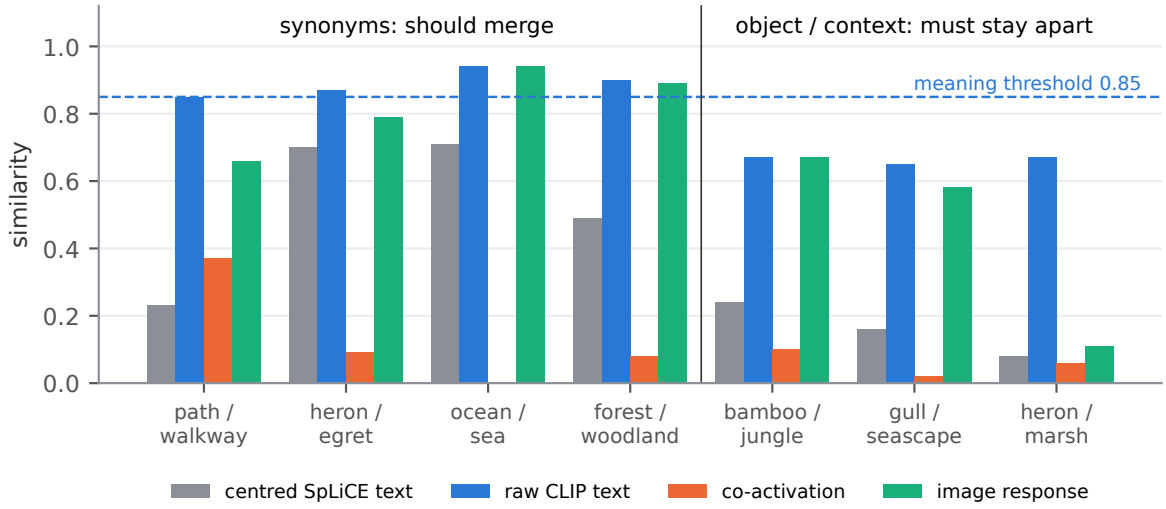


Figure 4: Four similarity signals on Waterbirds concept pairs. Co-activation (orange) fails for synonyms: the L1 penalty activates one of two synonyms per image (heron and egret co-activate at 0.09, ocean and sea at 0.00). Centred SpLiCE text directions (grey) also separate synonyms. Raw CLIP text embeddings (blue) put synonyms at 0.85 to 0.94 and object–context pairs at 0.64 to 0.72, so a threshold at 0.85 separates them.

The original CoSpRo rule linked two concepts when both centred text similarity and co-activation passed a threshold. On Waterbirds 499 of 502 groups were single concepts; lowering both thresholds chained a bird to its background (one group held hummingbird, parrot, jungle and bamboo). The **meaning grouping** (`cospro/cli/build_meaning_groups.py`) fixes both problems:

1. keep concepts active on 0.2% to 95% of the training images;
2. similarity $s_{k\ell} = \cos(t_k, t_\ell)$ of the raw CLIP text embeddings, set to -1 (veto) when the images' responses to the two concepts correlate below 0.5;
3. average-linkage clustering cut at text similarity 0.85. Average linkage joins two clusters only when their members agree on average, which stops chains;
4. a factor's text direction is the normalized mean of its members' text embeddings;
5. keep factors active on 2% to 90% of the images (5% to 90% for the image-level variants).

Examples: {kitty, kitten, cat}, {canine, dog}, {cozy, comfy, cosy}. Result: 242 factors on Waterbirds, 130 on MetaShift and 147 on Spur-CIFAR10 (2% band).

Why these numbers

- **Text 0.85**: lies between synonyms (0.85–0.94) and object–context pairs (0.64–0.72); a sweep over 576 factor sets on MetaShift chose it together with the response veto 0.5.
- **Response veto 0.5**: two words with similar text but images that respond differently stay apart (heron and marsh: response 0.11).
- **Frequency band 2%–90%**: a factor on fewer than 2% of the images occurs about once per half batch (64 images), so a fit on half a batch cannot learn it; at 5% the MetaShift factors lose every scene concept (couch, lawn, bed, window, bench), at 2% they keep them. Factors above 90% carry no contrast between images.

5 The month at a glance

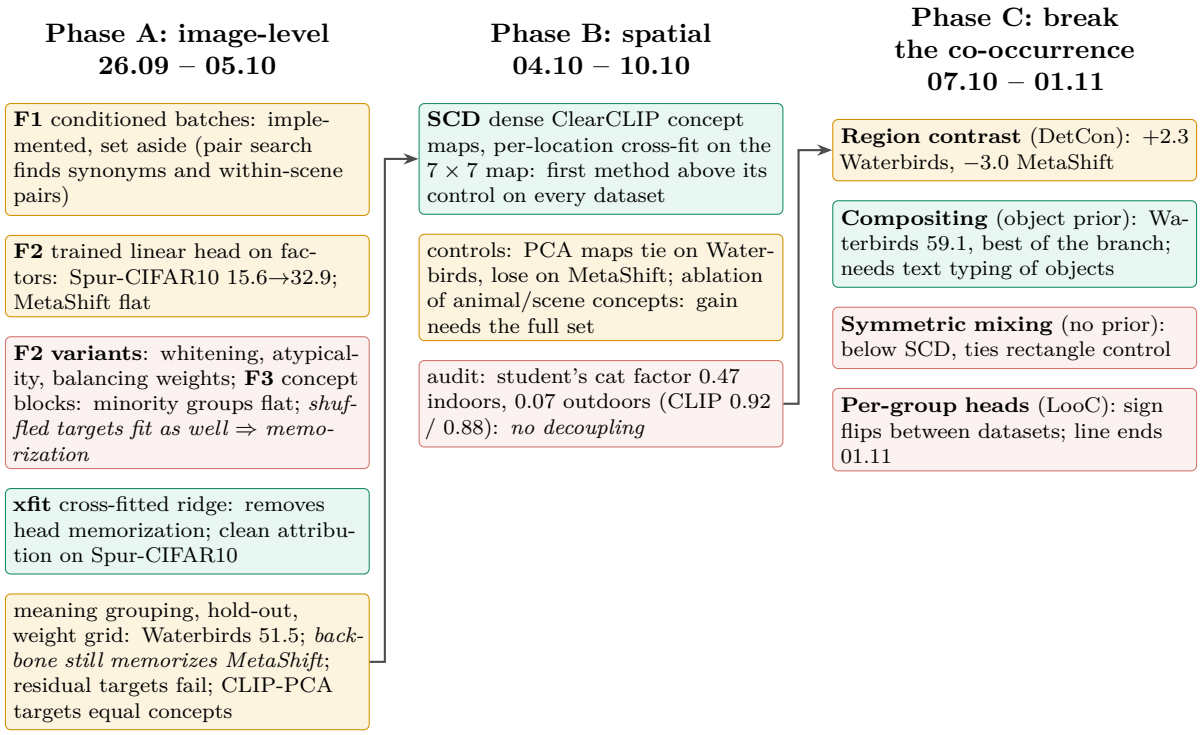


Figure 5: Roadmap of the branch. Green: worked and kept; yellow: partial or informative; red: failed. Each stage answered a question the previous one raised.

6 Phase A: image-level concept distillation

6.1 F1: concept-conditioned batches (set aside)

Idea. Pick pairs of factors whose presences correlate strongly across images (phi coefficient) and whose words name different things. With some probability fill a batch only with images that show one factor of a pair. Inside such a batch the factor tells no image apart, so InfoNCE must separate the images by everything else, the partner factor included. This is conditional contrastive learning (Tsai et al., 2021) with a discovered factor in place of an annotated attribute.

Why it was never trained. Ranking pairs by phi filled the top eight with two names for one

thing (cat breeds, lorry and truck). After the meaning grouping the top pairs were within-scene pairs (windshield and driving 0.64, toilet and sink 0.57) and the class–context pairs came last (dog and lawn 0.18, cat and couch 0.04): phi penalizes pairs of very different frequency (cat on 36% of the images, couch on 3%). Pair precision, the share of top pairs that join a class factor to an attribute factor, was zero. F2 and every later method use all factors and need no pairs, so F1 was set aside.

6.2 F2: factor distillation with a trained linear head

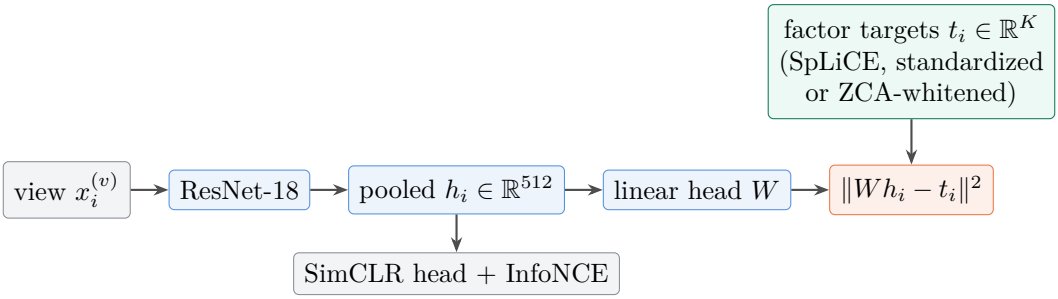


Figure 6: F2. A linear head predicts each image's factor values from the pooled backbone feature. The head is linear so the factors must be linearly readable, as the probe reads them.

Loss: $\mathcal{L} = \mathcal{L}_{\text{SimCLR}} + \lambda \frac{1}{2B} \sum_{i,v} \|W h_i^{(v)} - t_i\|^2 / K$, weight λ switched on at epoch 10 with a 10-epoch linear warm-up. Whitened targets (ZCA, ridge 0.1) keep for each factor the part the other factors leave unexplained, which is large exactly on the images that break a correlation.

Table 4: F2 rounds (validation WGA / average accuracy; groups on MetaShift: cat-in / cat-out / dog-in / dog-out).

Dataset	Arm	WGA	Acc.	Groups
Spur-CIFAR10	SimCLR	15.6 ± 1.0	59.5	
	F2, standardized	32.9 ± 1.6	69.5	
	F2, shuffled targets	24.2 ± 0.4	58.6	
MetaShift	SimCLR	44.3 ± 7.3	55.2	57 / 47 / 53 / 64
	F2, weight 3	46.6 ± 5.7	61.5	70 / 47 / 54 / 74
	F2, weight 3, shuffled	43.3 ± 8.1	54.8	57 / 45 / 53 / 65
	F2, whitened, weight 3	47.9 ± 6.4	56.9	57 / 51 / 55 / 65
	F2, atypicality weights	50.5 ± 3.8	58.8	59 / 52 / 55 / 70
	F2, balancing weights	44.2 ± 8.1	59.0	64 / 44 / 55 / 72
Waterbirds	F2 (frozen from round 1)	41.6 ± 6.1	51.2	below SimCLR

Atypicality weights scale each image's loss by its mean squared whitened factor value (how strongly its concepts break the dataset's correlations; minority images receive about twice the weight on a synthetic fixture). **Balancing weights** are per-image weights that minimize the squared correlations between factor presences, so that under them "indoor" no longer predicts "cat" (stable learning, Zhang et al., 2021).

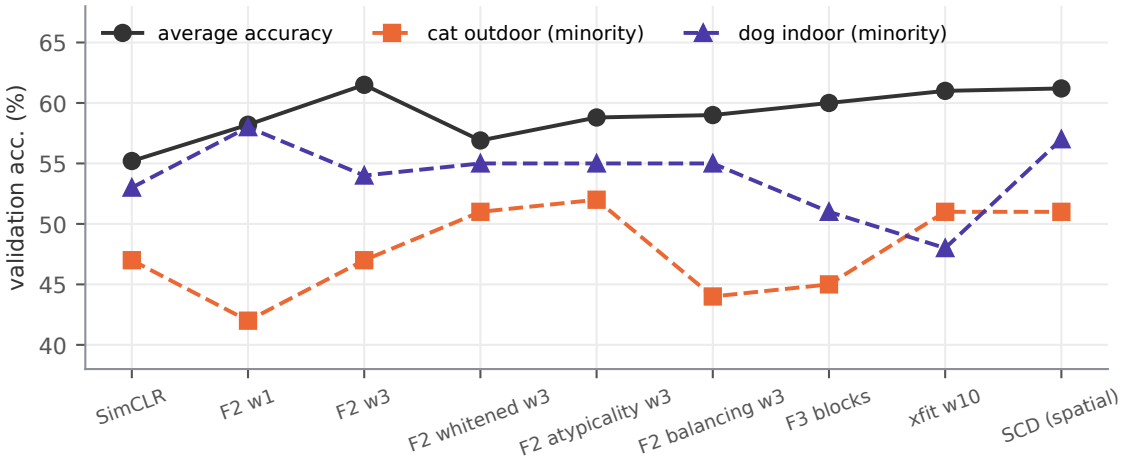


Figure 7: MetaShift through the image-level rounds. Average accuracy rises with almost every concept loss; the two minority groups (cat outdoors, dog indoors) stay within a few points of SimCLR. The gain goes to the majority groups.

6.3 F3: concept blocks (supervised contrast per factor)

One small linear block per factor (32 most balanced factors). Inside block k, images that both show factor k are positives (SupCon) and a positive pair weighs more when the two images' other factors differ ("cat on sofa" with "cat on street"). A fused "cat-and-indoor" direction cannot rank "cat on street" above "dog on sofa" inside the cat block, so the backbone would need a cat direction that survives the context. Result: 43.8 ± 4.1 WGA; its control with uniform pair weights scored 49.7 ± 2.5 and its shuffled control 46.4 ± 3.5.

Verdict on F2 and F3: memorization

The controls explained every MetaShift failure. F2 at weight 3 explained 0.41 of the real target variance and **0.33 of the shuffled one**; the block loss ended at 3.34 with real concepts and 3.37 with shuffled ones. On 1,700 images the network fits a per-image target through the features SimCLR learns anyway to tell images apart and random targets are fitted almost as well as real ones (as Zhang et al. 2017 showed for labels). Any auxiliary loss attached to single images is then met without learning concepts. On Spur-CIFAR10 (45,000 images) shuffled targets reach only 0.02, so the concepts shape the student there.

6.4 xfit: the cross-fitted ridge

Idea of the cross-fit

Replace the trained head by a ridge regression *fitted in closed form on one half of the batch and evaluated on the other half.* A feature that only identifies one image carries nothing from half A to half B, so memorizing images lowers no part of the loss; only concept directions shared across images do. This is the sample splitting of double machine learning applied to a training loss, implemented like the differentiable closed-form heads of R2D2 and MetaOptNet.

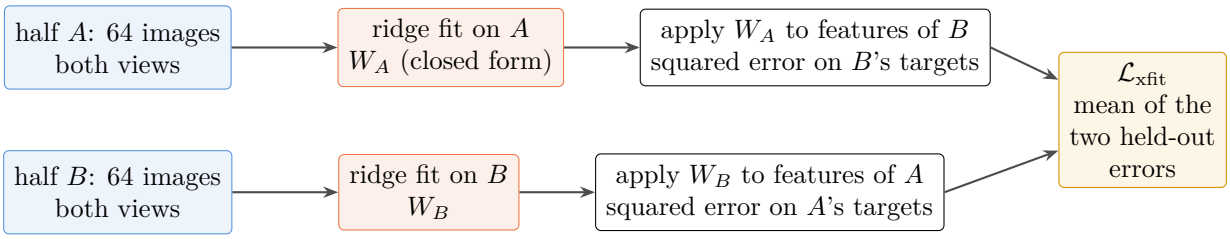


Figure 8: Cross-fitting. Gradients flow through the solve and the prediction into the backbone. Both views of an image stay in the same half, so no half sees a view of an image it predicts.

With unit-norm centred features H, targets T and per-image weights D, the kernel form is

$$\hat{T}_B = H_B H_A^\top \left(D_A H_A H_A^\top + \rho I\right)^{-1} D_A T_A, \tag{3}$$

one $n \times n$ solve per half (code: `cross_fit_predictions` in `cospro/methods/concept_factors.py`).

Why unit-norm rows and an absolute ridge $\rho = 1$

The first version scaled the ridge with the feature norm and was small. Fits of 64 images in 512 dimensions nearly interpolated, the solve became ill conditioned, three of four Spur-CIFAR10 runs produced NaN features and the held-out explained variance was negative everywhere. Normalizing every feature row stops the backbone from weakening the ridge by shrinking its features and an absolute ridge keeps every eigenvalue of the system at least ρ. Ridge 10 performed no better on MetaShift (42.5 WGA).

Table 5: Repaired cross-fit (seeds 1 and 2). Batch EV: held-out explained variance of the in-batch fit.

Dataset	Arm	WGA	Acc.	Batch EV
Spur-CIFAR10	SimCLR	15.6 ± 1.0	59.5	
	F2 trained head	32.9 ± 1.6	69.5	0.10
	xfit	33.6 ± 2.3	69.7	0.13
	xfit, shuffled	18.2 ± 3.2	57.5	-0.02
MetaShift	xfit (145 factors)	43.9 ± 3.2	54.8	-0.01
	xfit, shuffled	41.7 ± 0.0	54.4	-0.03

On Spur-CIFAR10 the cross-fit matches F2 and its shuffled control falls to the SimCLR level: the **whole gain comes from the concept content**, the first clean attribution of the branch.

6.5 What the student can learn: bounds, weights and unseen images

Learnability bound. A ridge fitted on 64 images can only predict factors that occur often enough. Running the same fit on frozen CLIP features gives an upper bound: with 278 factors from 1% frequency even CLIP scores -0.015; with the 24 factors from 5% it scores 0.16 (shuffled: -0.07). Hence the 5% band for image-level runs.

Loss weight. At weight 10 the 24 MetaShift factors reached 0.72 to 0.87 explained variance on the training images, *three times the CLIP score*, while WGA stayed at 45.5. A probe on the 24 factor values themselves reaches 77% WGA. The student stored something other than the concepts.

The hold-out diagnostic. A fixed fifth of the training images gets loss weight 0 in the concept loss (SimCLR still trains on them). A ridge fitted on the other images then scores the factors on these *unseen* images. This exposes memorization in the backbone, which the cross-fit cannot see.

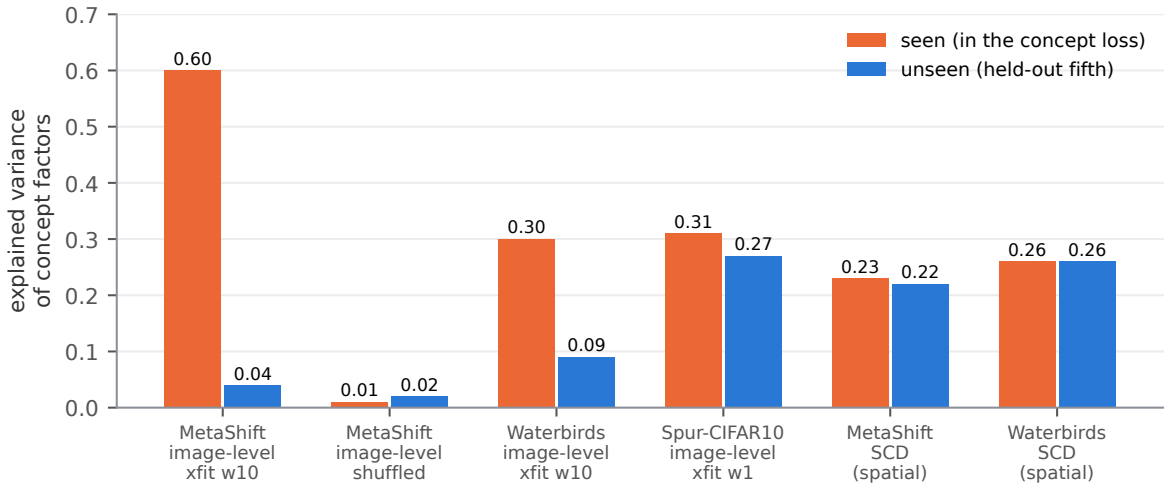


Figure 9: Explained variance of the factors on images inside the concept loss (seen) and outside it (unseen). Image-level xfit on MetaShift: 0.60 seen, 0.04 unseen, the backbone stores the targets of its 1,700 images. Spur-CIFAR10 (45,000 images) generalizes: 0.31 against 0.27. SCD (spatial, scored on dense responses) shows no gap on either dataset.

6.6 Waterbirds with the frozen image-level configuration

The configuration chosen on MetaShift (meaning factors from 5%, 70 Open Images factors, weight 10, hold-out) reached 55.3 WGA on the first two Waterbirds seeds and 51.5 ± 5.5 on four, with accuracy 57.5 against 52.3 for SimCLR and 41.4 ± 3.0 for its shuffled control. The same configuration on Spur-CIFAR10 dropped to 22.7 (weight 1 gave 33.6): at weight 10 even 45,000 images are memorized. The weight grid on Spur-CIFAR10 showed that the unseen score cannot choose the weight label-free (0.32 to 0.33 for weights 1 to 10 while WGA moved within noise).

6.7 Two diagnostics that ended Phase A

Decoupling score. Each factor minus its linear regression on the other factors (its *residual*) is the part a fused direction cannot predict. The student explains 0.03 to 0.04 of the residuals against 0.15 to 0.17 for CLIP. Training directly on residual targets failed: the student could not predict them and Waterbirds WGA fell to 41.7 ± 0.9, the level of the shuffled control.

CLIP control. Distilling as many principal components of the CLIP image embeddings as there are factors gives the same or better numbers (Waterbirds 51.0, MetaShift 54.2, Spur-CIFAR10 36.3 to 37.0). At the image level the concepts add nothing over generic CLIP information.

Verdict on Phase A

Image-level concept distillation helps when the student can generalize the targets (Spur-CIFAR10: 15.6 to 33.6; Waterbirds: 46.7 to 51.5). Cross-fitting makes the attribution clean. But (i) small datasets are memorized in the backbone, (ii) the student learns the part the factors share, i.e. the dataset's co-occurrence and (iii) named concepts are not better than CLIP principal components. Image-level targets can separate "cat" from "couch" only through the images where the two disagree; that is the dead end that motivated the spatial loss.

7 Phase B: Spatial Concept Distillation (SCD), the main method

Idea of SCD

At the image level cat and couch appear together; at the pixel level they occupy different places in every image. So tell the student *where* each concept is: a frozen CLIP model computes a dense map of every concept factor over every training image and every location of the student's 7×7 feature map must predict the concept values at its place in the view, through the same cross-fitted ridge.

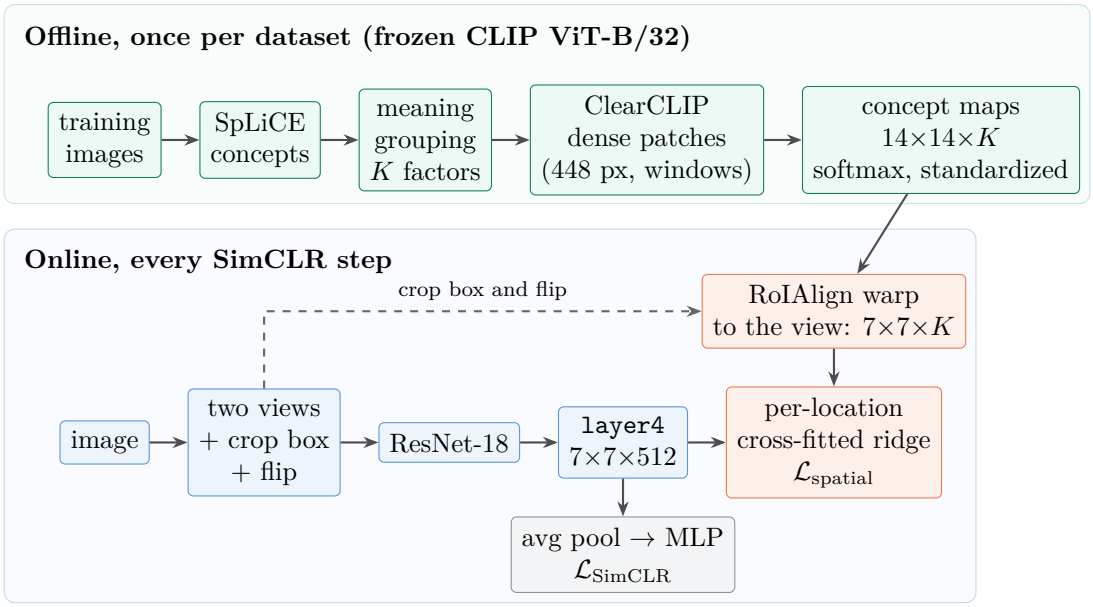


Figure 10: SCD. The teacher branch runs once (40 seconds on one GPU for MetaShift) and stores the maps. The student branch adds one loss to unchanged SimCLR. At inference only the student backbone and the probe remain; teacher, maps and heads are discarded.

7.1 Step 1: dense concept maps from CLIP

Plain CLIP patch tokens of the last layer do not align with text, because the final attention pools global information into every token. ClearCLIP (Lan et al., ECCV 2024) fixes this in the last transformer block: query–query attention in place of query–key attention and no residual and feed-forward branches. With X the tokens entering that block,

$$P = \text{norm}\Big(\text{LN}\big(\text{softmax}(QQ^\top/\sqrt{d_h})\,VW_O\big)W_{\text{proj}}\Big), \qquad Q = XW_Q,\ V = XW_V. \tag{4}$$

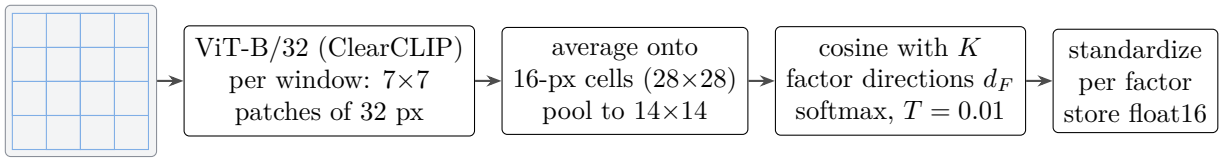


image at 448 px, 9 windows of 224 px, stride 112

Figure 11: Map construction. Code: `cospro/pipeline/concept_maps.py`; launcher: `scripts/build_concept_maps.sbatch`.

At every cell u the map is a softmax over the factors,

$$M_{iF}(u) = \frac{\exp(\cos(p_i(u), d_F)/T)}{\sum_G \exp(\cos(p_i(u), d_G)/T)}, \qquad \tilde{M}_{iF} = (M_{iF} - m_F)/s_F. \tag{5}$$

Why 448 pixels, windows, 14×14 and $T = 0.01$

- **448 px through 224-px windows.** ViT-B/32 sees 224-pixel inputs as only 7×7 patches of 32 pixels, too coarse to separate a cat from its cushion. Nine overlapping windows (stride 112) at 448 pixels give an effective 16-pixel grid while each window stays at the resolution CLIP was trained on.
- **14×14, twice the student's 7×7.** A SimCLR crop covers 20% to 100% of the image; a 20% crop still spans about 6×6 map cells, so RoIAlign has real content to resample into the 7×7 target.
- **Softmax with $T = 0.01$.** Cosines of a patch with different text directions differ by a few hundredths; $T = 0.01$ turns them into a peaked distribution, so each cell distributes one unit of mass over the factors and mostly picks a winner.
- **Standardized per factor.** Every factor then has unit variance, so frequent factors (blanket, lawn) do not dominate the squared error.

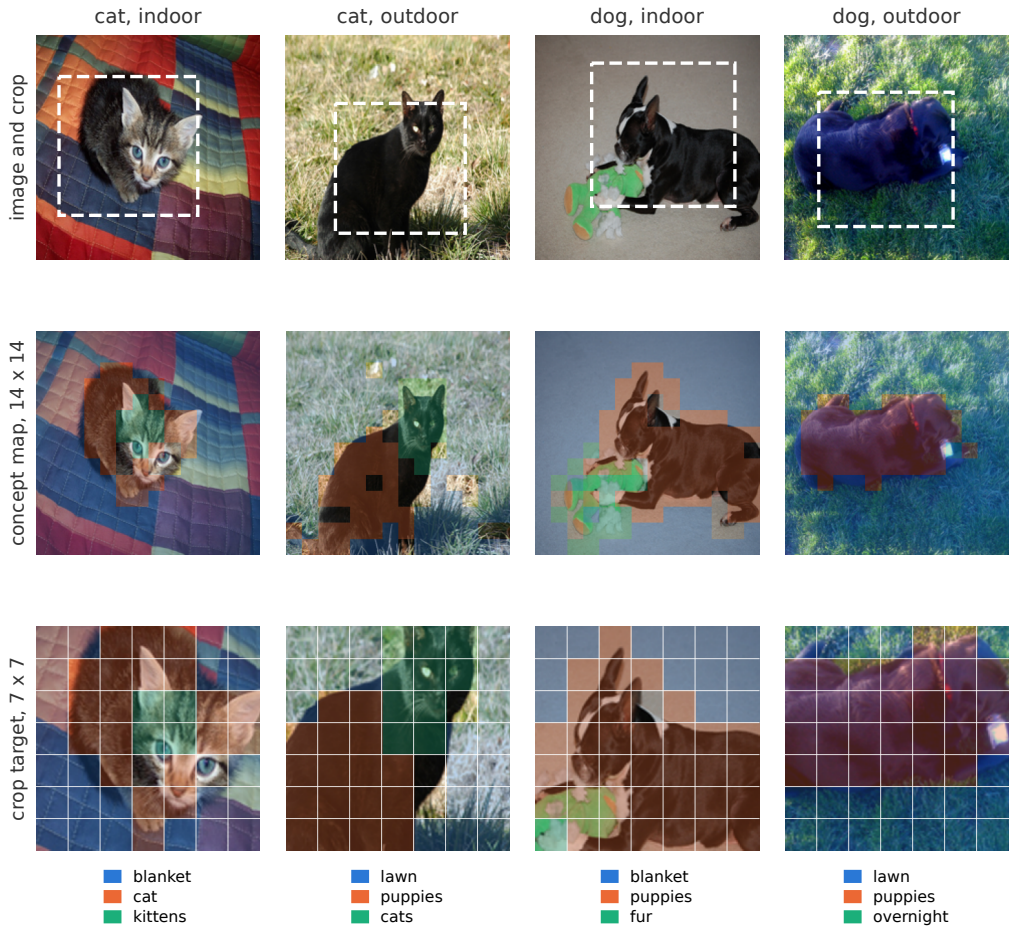


Figure 12: MetaShift concept maps. Top: image and one SimCLR crop (dashed). Middle: the winning factor of each of the 14×14 cells (the three most frequent factors coloured). Bottom: the crop's 7×7 targets after warping. The animal factors cover the animal, the scene factors its surroundings. The teacher is imperfect: in the cat-outdoor image part of the black cat is labelled "puppies".

7.2 Step 2: warping the maps onto each view

The SSL transform records the crop box and flip of every view. RoIAlign cuts the view's box out of $\tilde{M}_i$ (a 2×2 sampling grid per output bin), resizes it to 7×7 and flips it with the view. Each view thus gets targets $Y_i^{(v)} \in \mathbb{R}^{K \times 7 \times 7}$ that describe exactly the pixels it shows.

7.3 Step 3: the per-location cross-fitted ridge

Every location u of the feature map gives a unit-norm row $r = \mathrm{norm}(H_i^{(v)}(u)) \in \mathbb{R}^{512}$ and a target $y = Y_i^{(v)}(u) \in \mathbb{R}^K$. The batch splits into halves A and B (both views of an image in one half). The ridge is fitted on all locations of A,

$$W_A = (\bar{R}_A^\top \Omega_A \bar{R}_A + \rho I)^{-1} \bar{R}_A^\top \Omega_A (Y_A - \bar{y}_A), \tag{6}$$

predicts every location of B and vice versa. Here the primal form is cheaper: one 512×512 solve over $64 \cdot 2 \cdot 49 = 6{,}272$ rows. The loss is the held-out squared error, averaged over both directions; it runs in float32 outside autocast. The full objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{SimCLR}} + \lambda(t)\, \mathcal{L}_{\mathrm{spatial}}, \qquad \lambda(t) = 0 \text{ until epoch 10, linear to } \lambda_{\max} = 5 \text{ at epoch 20.} \tag{7}$$

Why these SCD settings

- **Cross-fit instead of a trained head**: the lesson of Phase A; a trained head stores per-image targets on small datasets.
- $\lambda_{\max} = 5$: from the weight grid 1, 3, 5 and 10 of the image-level loss; spatial weight 10 adds nothing (Waterbirds 53.5, MetaShift 48.3, two seeds).
- **Start at epoch 10, warm-up 10 epochs**: SimCLR forms features before the concept loss acts; inherited from F2, where starting at epoch 0 changed nothing.
- **20% hold-out**: introduced as a diagnostic (unseen images); the arm with the hold-out scored higher than the arm without it (Waterbirds 52.8 vs. 49.5, MetaShift 51.1 vs. 46.7, six seeds) and became the main arm. The difference lies within the seed spread and has no explanation yet.
- **Batch 128**: 6,272 rows per half against 512 dimensions keeps the fit well posed.

7.4 Results

Table 6: SCD against SimCLR and the shuffled-map control. Waterbirds and MetaShift: seeds 1–6; Spur-CIFAR10: seeds 1–2. Validation, mean of the last four probes.

	Waterbirds		MetaShift		Spur-CIFAR10	
Method	Avg.	WGA	Avg.	WGA	Avg.	WGA
SimCLR	52.4	46.7 ± 1.9	55.2	44.3 ± 7.3	59.6	15.6 ± 1.0
Shuffled maps	52.8	47.5 ± 2.5	56.1	45.1 ± 2.9	55.9	17.3 ± 2.0
SCD	**60.3**	52.8 ± 4.7	**61.2**	51.1 ± 5.4	**69.3**	38.1 ± 0.5

SCD beats SimCLR by 6.1 (Waterbirds), 6.8 (MetaShift) and 22.5 (Spur-CIFAR10) points of WGA and by 7.9, 6.0 and 9.8 points of average accuracy. It is ahead on five of six paired seeds on Waterbirds and MetaShift and on both Spur-CIFAR10 seeds. The shuffled control stays within one point of SimCLR on Waterbirds and MetaShift and 1.6 points on Spur-CIFAR10, so the gain comes from the content of each image's own maps.

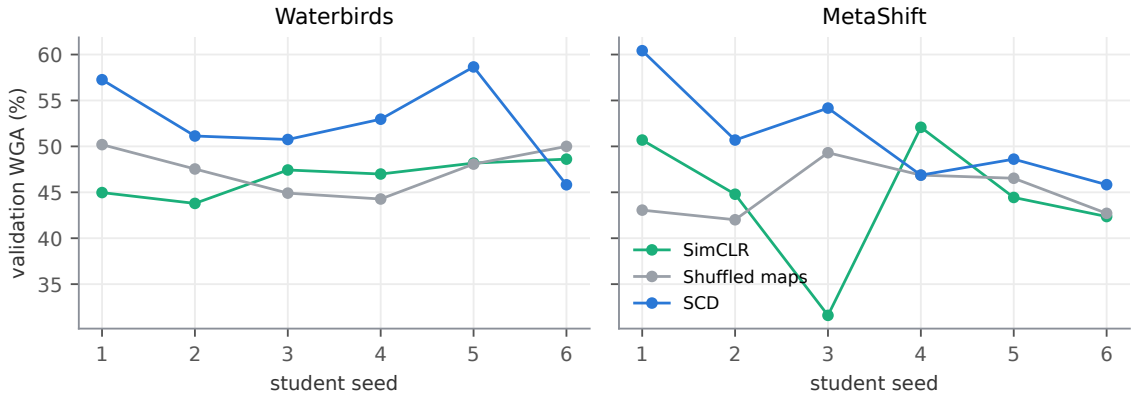


Figure 13: Per-seed WGA. MetaShift seeds spread by more than 20 points for SimCLR (31.6 to 52.1), which is why we report six seeds and paired differences. Seed 6 is hard for every method on Waterbirds.

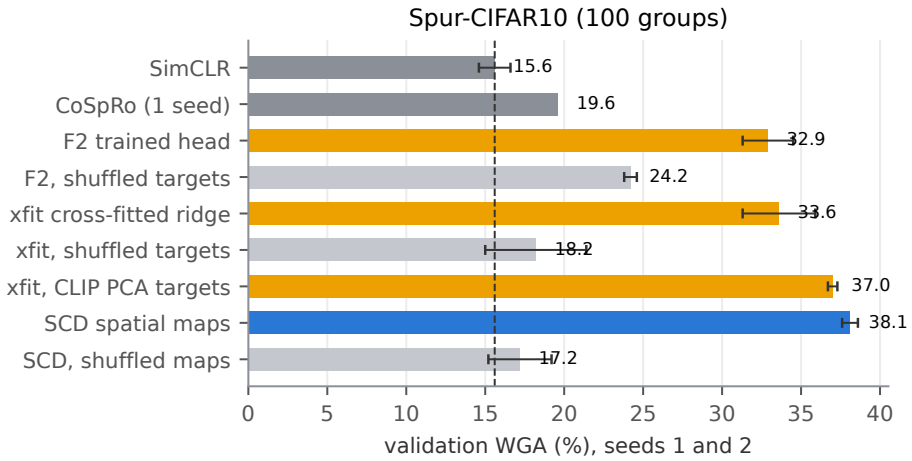


Figure 14: Spur-CIFAR10, where the spurious feature is a thin coloured line. Every real-target method roughly doubles WGA and every shuffled control stays near SimCLR. SCD gives the best result of the branch.

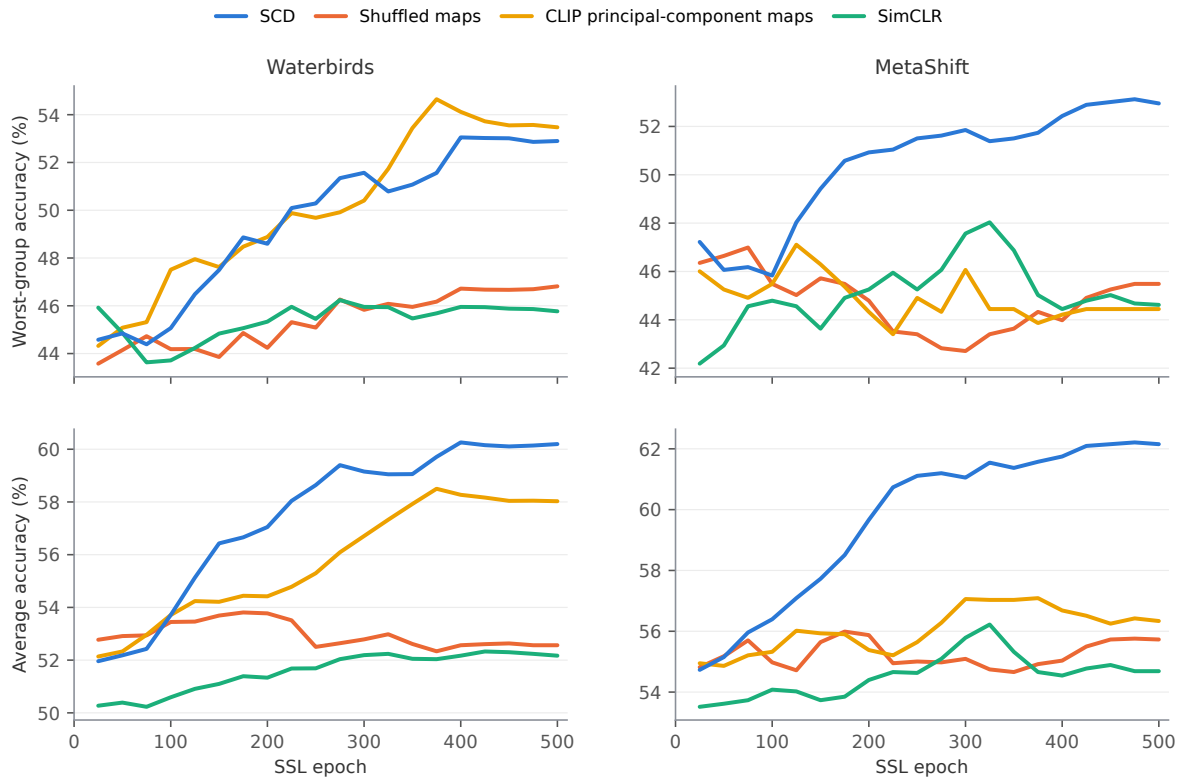


Figure 15: Validation accuracy during SSL training (seeds 1–4, 3-point smoothing). On MetaShift SCD leads every method from epoch 100 onward; on Waterbirds it leads SimCLR and the shuffled control from epoch 100 and the PCA-map control follows the same path.

7.5 Controls: is it the concepts, CLIP or the location?

Table 7: Controls of SCD, seeds 1–4 (image-level row on MetaShift: seeds 1–2).

	Waterbirds		MetaShift	
Method	Avg.	WGA	Avg.	WGA
SimCLR	52.3	45.8 ± 1.7	54.8	44.8 ± 9.4
Shuffled maps	52.6	46.7 ± 2.7	55.6	45.3 ± 3.4
CLIP principal-component maps	58.1	53.6 ± 1.9	56.3	44.4 ± 7.4
Image-level factor regression	58.0	50.4 ± 9.0	61.7	47.2 ± 3.9
SCD	60.2	53.0 ± 3.0	62.2	53.0 ± 5.8

- **PCA maps** keep the dense CLIP information of the teacher and drop the concept names (as many principal components of the ClearCLIP patch embeddings as there are factors). On MetaShift SCD wins on all four seeds ($+16.0, +1.4, +4.2, +12.9$); the PCA maps stay at the SimCLR level. On Waterbirds the two tie: birds against land and water are among the strongest directions of CLIP's patch space, so the leading components carry them; cats against dogs and couches against lawns are weaker directions that only the named concepts single out.
- **Image-level regression** keeps the concepts and drops their locations; SCD is ahead by 2.7 WGA on Waterbirds, 8.3 on the two paired MetaShift seeds and 8.7 on Spur-CIFAR10.
- **Concept ablation** on MetaShift (seeds 1–4): without the 21 animal factors 46.6 ± 5.8, without the 20 scene factors 47.9 ± 2.8, all factors 53.0 ± 5.8. Both ablations lose most of the gain: it needs the full concept set.

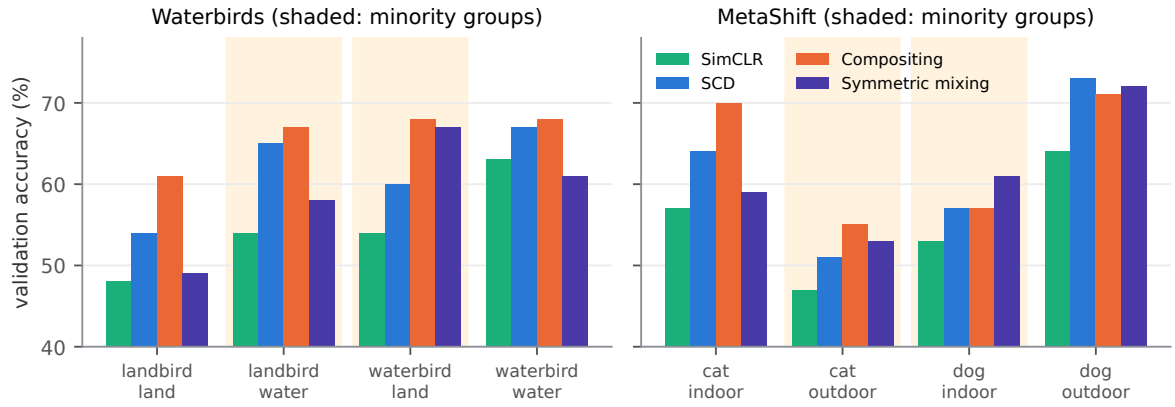


Figure 16: Accuracy per group (six seeds for SimCLR and SCD; compositing six seeds on Waterbirds, four on MetaShift; mixing four). SCD raises every group, the minority groups included; compositing flattens the Waterbirds groups at 61 to 68.

7.6 What the student learns: the audit

It generalizes. On the held-out fifth the student's pooled features explain 0.22 (MetaShift) and 0.26 (Waterbirds) of the image-level factor responses, against 0.13 and 0.17 with shuffled maps; scores on images inside the loss agree (0.23 and 0.26). The memorization of Phase A is gone (Figure 9).

It does not decouple. With labels used only for this audit, the student's *cat* factor scores 0.47 on unseen cats indoors and 0.07 on unseen cats outdoors; CLIP scores 0.92 and 0.88 (Figure 17). The factor residuals score 0.00 in the student against 0.15 to 0.17 for CLIP. A background probe on the frozen features reaches 90.5% WGA for SCD against 89.5% for SimCLR: SCD erases no background.

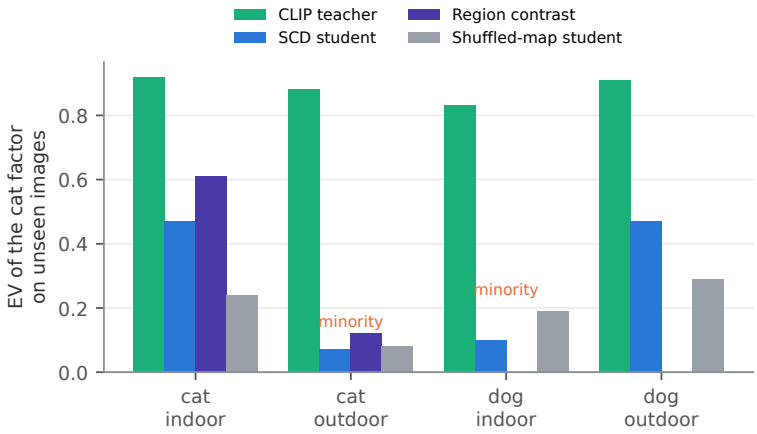


Figure 17: How well the *cat* factor is predicted on unseen MetaShift images, per group. CLIP sees cats everywhere; every student sees cats where cat and scene agree (cat indoors) and fails on the minority groups.

Verdict on SCD

SCD is the first method of the project that beats its own control on every dataset tested and generalizes its concepts to unseen images. It raises WGA because the balanced probe gains a usable core feature that SimCLR from scratch lacks; the spurious attribute stays as readable as before. The student still predicts "cat here" from local evidence *plus* scene, because each 7×7 location sees most of the view and the scene agrees with the cat on 88% of the training images. This is the prior-free method of the thesis.

8 Phase C: making the student separate object and context

Every loss so far is an expectation over the natural training images, where cats sit indoors. Phase C tried to change that distribution or to demand that a concept keeps its identity in other surroundings. All four variants run on top of SCD (weight 5, 20% hold-out).

8.1 Region contrast (DetCon-style)

Idea. Pool each view's feature map inside its two dominant concept regions (warped map probability as weight, mean probability at least 0.1), project with a small head and contrast regions across the batch: regions of the same concept in other images are positives, weighted towards pairs whose images' concept mixes differ. A cat region on a lawn must resemble a cat region on a couch. About 335 regions per MetaShift batch, 94% of them with positives.

Verdict

Waterbirds 55.3 ± 3.1 (SCD 53.0), MetaShift 50.0 ± 6.5 (SCD 51.1, below on three of four seeds). The concept scores on unseen images rise on every seed (MetaShift 0.22 $\rightarrow$ 0.29) and the cat factor on cats outdoors from 0.05 to 0.12, yet the probe gain goes to the majority groups. On Waterbirds the shuffled region arm gains as much over shuffled SCD as the real one over SCD: part of the gain is region pooling itself.

8.2 Concept compositing (with the object prior)

Idea. Change the training distribution itself. For 30% of the qualifying images, cut the *object* region out of the second view and paste it onto another image's view whose own objects are painted over (normalized convolution from their surroundings). The composite replaces the second SimCLR view, and its concept targets are composited with the same mask. With a random donor, object and context become independent in the composites: they sample the interventional distribution in which the context carries no information about the object, the content–style condition under which contrastive learning isolates content (von Kügelgen et al., NeurIPS 2021).

Which concepts are "objects" is decided by text alone (`cospro/pipeline/concept_type.py`): CLIP text similarity to scene prompts minus similarity to object prompts; factors below the median count as objects. No label enters, but the step *types* concepts. On MetaShift the 65 object factors hold every animal and breed (and also verbs and people); lawn, couch, bed, desk, window, beach and snow are context.

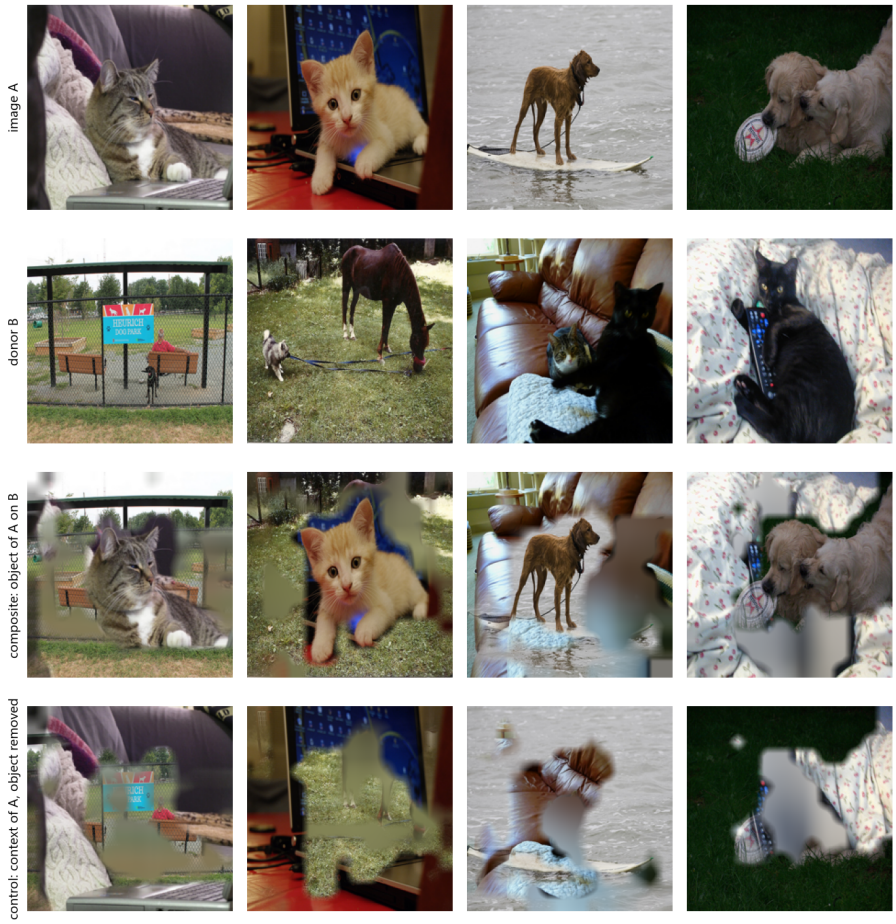


Figure 18: Real MetaShift composites. Row 3: the object of image A on donor B (the training input). Row 4: the "context kept" control, which pastes A's surroundings instead. Seams are half a map cell wide.

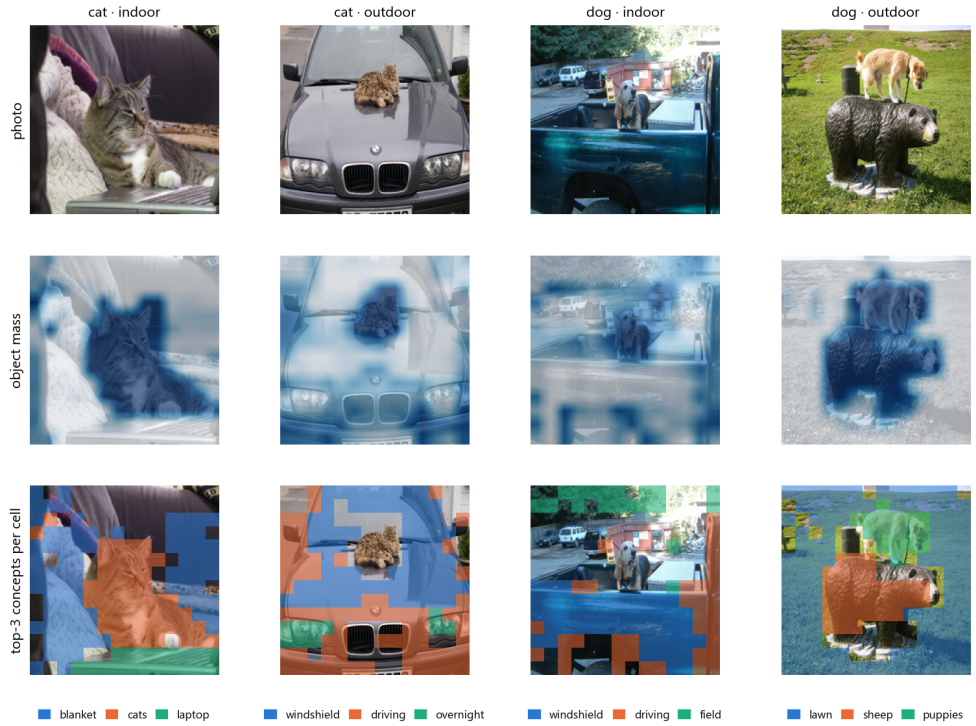


Figure 19: The object mass that compositing cuts (row 2) and the three leading concepts per cell (row 3) on one MetaShift image per group. The teacher often names the context well (windshield, driving, lawn) and the animal in broad strokes.

Table 8: Compositing and its controls (validation WGA / Avg.). Waterbirds seeds 1–6 for compositing and the silhouette control, 1–4 otherwise; MetaShift seeds 1–4.

Arm	Waterbirds	MetaShift
SCD	52.8 / 60.3	51.1 / 61.2
Compositing, object regions	59.1 ± 7.1 / 64.7	52.3 ± 1.9 / 63.3
control: another image's object silhouette	54.2 ± 6.1 / 62.5	50.6 ± 3.7 / 60.0
control: context region pasted instead	52.7 ± 2.1 / 60.0	49.3 ± 1.9 / 60.1
control: shuffled maps	52.9 ± 1.0 / 57.2	48.4 ± 3.1 / 56.6

Verdict: strongest result, but it uses a prior

On Waterbirds compositing beats SCD on all six seeds (6.3 WGA on average), the context and shuffled controls on all four paired seeds and the silhouette control on four of six; its groups flatten to 61–68. On MetaShift it gives the steadiest WGA of the branch (sd 1.9). The arms order as the concept hypothesis predicts: object regions > random silhouettes > context regions. The gain rests on the object prior (target = object, spurious attribute = surroundings), which the restriction excludes and which CelebA breaks. Compositing therefore serves as the measure of what a correct prior adds.

8.3 Symmetric concept mixing (no prior)

Idea. Treat every concept alike. Keep each concept region of view A with probability one half (kept area 25–75%), fill the rest from a random donor B: $x = M \odot A + (1 - M) \odot B$. The mixed view is a third encoder pass. Its feature map pooled inside M is a positive of A, pooled outside a positive of B (RegionCL), and its global embedding is a soft positive of A and B in proportion to their areas

(Un-Mix). Under mixing the twelve strongest animal-by-scene correlations on MetaShift fall from mean $|\phi| = 0.33$ to 0.19 (object pasting: 0.17).

Verdict

Waterbirds 49.2 ± 2.5 (below SCD on all four seeds), MetaShift 51.0 ± 8.5; it ties its rectangle control on both datasets (49.0 and 51.8), so concept boundaries make no difference. Typing no concept leaves an augmentation that behaves like CutMix and weakens the majority group landbird-on-land (49 vs. 55).

8.4 Per-group projection heads (LooC-style, no prior)

Idea. Symmetric mixing asked one embedding to stay the same whatever part was replaced, sometimes the scene and sometimes the cat; the demands may cancel. LooC (Xiao et al., ICLR 2021) gives each invariance its own embedding space. Factors are clustered into 12 groups by the pixels they share in the maps (spectral clustering of map co-activation, no label or prompt). For half of the images one group is kept on a donor; the mix is a positive of its image only in that group's own projection head.

Verdict

Waterbirds 52.4 ± 2.9 (ties SCD, beats the single-head control by 9.5); MetaShift 46.9 ± 3.3 (loses to the single head by 5.2, to SCD by 6.2 and to its shuffled control). The head effect flips sign between datasets. The heads trained well (highest unseen concept scores of all arms, 0.28 and 0.32), so concept scores do not predict WGA. The cat and dog groups were kept in only 21% of the mixes: a draw that favours the class needs to know the class, which is the prior again. The prior-free mixing line ended on 1 November.

9 The scoreboard, row by row

Table 9: All architectures with their idea, outcome and the lesson they taught. WGA on Waterbirds / MetaShift (validation).

Architecture	Idea	WGA (WB / MS)	Lesson
F1 batches	condition batches on one factor of an entangled pair	not trained	phi ranks synonyms and within-scene pairs first
F2 trained head	regress factor values from the pooled feature	41.6 / 46.6	per-image targets are memorized on small datasets
F3 concept blocks	SupCon per factor, cross-context positives	– / 43.8	shuffled concept sets fit as well
xfit	ridge fitted on one half, tested on the other	51.5 / 47.6	removes head memorization; backbone still memorizes; CLIP PCA as good
SCD	per-location cross-fit on dense concept maps	**52.8 / 51.1**	location helps; generalizes; does not decouple
Region contrast	DetCon over concept regions	55.3 / 50.0	raises concept scores, not minority groups
Compositing	paste object on another scene	**59.1 / 52.3**	works, needs object-vs-scene prior
Symmetric mixing	keep random concept regions, soft positives	49.2 / 51.0	equals CutMix with rectangles
Per-group heads	one embedding space per concept group	52.4 / 46.9	sign flips; drawing the class needs the prior

10 What to say about the bigger picture

1. **Concepts help a from-scratch SSL student when they are delivered where they are.** Image-level distillation is limited by memorization and by co-occurrence; spatial distillation generalizes and lifts every group.
2. **Every claim stands on a matched control.** Shuffled targets for content, PCA maps for named concepts versus generic CLIP, image-level regression for location, ablations for which concepts.
3. **The core feature becomes stronger, but it is not separated from its context.** DFR needs separate directions; SCD supplies a stronger core direction, still fused with the scene. Breaking the fusion worked only with an object prior.
4. **The decision for the thesis.** The prior-free method is SCD. Compositing reports what a correct assumption about the spurious attribute adds, stated openly, as the background-augmentation literature does.

11 Limitations and open items

- **Validation only.** `FINAL_TEST=1 bash scripts/submit_factor_study.sh <dataset>` runs the same arms once on the locked test split (Waterbirds test: 642 images per minority group against 133 in validation); no test results are in the repository yet. The SCD settings (weight 5, hold-out) were chosen on validation of MetaShift and Waterbirds seeds 1–4; seeds 5 and 6 ran with the settings fixed.
- **Noise.** MetaShift's seed spread (5.4 WGA) is as large as the gain on single seeds.
- **The hold-out effect** (higher WGA with 20% of images out of the concept loss) has no explanation.
- **LateTVG matched comparison**: one seed under our protocol (47.9); the six-seed test run is predeclared.
- **CelebA**: concept maps for 162,770 images do not fit the loader's float32 copies in 40 GB; image-level xfit gives 85.9 against 84.9 for SimCLR.
- **Teacher limits**: CLIP's vocabulary and 32-pixel patches; Spur-CIFAR10 images are upsampled from 32 to 448 pixels for the maps.
- **Infrastructure**: several jobs stalled without an error on dgx nodes; at most eight jobs run at once.

Proposed next steps (to agree on in the meeting):

1. Locked test for SimCLR, shuffled maps, SCD and compositing on Waterbirds and MetaShift, six seeds.
2. Matched LateTVG on Waterbirds (predeclared) and MetaShift (prune rate chosen on validation first).
3. CelebA concept maps with a memory-lean loader, then SCD on CelebA as the held-out dataset whose spurious attribute is no background.
4. Decide how the thesis frames compositing: as an ablation that quantifies the object prior.

12 Questions the supervisor may ask

Q: Why train from scratch and use ResNet-18 instead of a pretrained model?

A: The protocol is inherited from CoSpRo and SpurSSL-style studies so that every method sees the

same images and budget. A pretrained backbone already encodes birds and cats from ImageNet and would mask what the SSL objective learns from the dataset. The cost is low absolute accuracy (52–61%).

Q: Is this not just CLIP distillation?

A: At the image level it is: CLIP principal components match the concepts. At the spatial level the concept maps beat PCA maps of the same width on MetaShift on all four seeds and tie on Waterbirds, where bird-versus-background is already a leading CLIP direction.

Q: Where do labels enter?

A: Only in the group-balanced probe (224 / 400 / 1,400 images) and the post-hoc audit, identically for every method. Concept discovery, grouping, maps and SSL use none. Hyperparameters were chosen on validation of MetaShift and Waterbirds, which is the main caveat.

Q: Why not use the pooled feature for the concept loss?

A: Pooling fuses co-occurring concepts into one vector; at 7×7 they occupy different locations (Section 4.2).

Q: Why does SCD not separate cat from indoor?

A: Each 7×7 location's receptive field covers most of the view and the scene predicts the cat on 88% of the training images, so the cheapest predictor of "cat here" uses the scene. The loss localizes concepts but keeps the training joint distribution.

Q: Why is compositing not the main method?

A: It needs a text prior that types concepts as objects or scenes. That prior holds for Waterbirds and MetaShift and fails for CelebA; the thesis restriction excludes it.

Q: How do you know the gain is not noise?

A: Six seeds, paired differences (SCD ahead on five of six on both datasets) and controls that share every statistic but the content, which stay at the SimCLR level.

Q: Why the last four probes?

A: One MetaShift validation image is 1.4 points of a group's accuracy; averaging the probes at epochs 425–500 damps that noise. For SCD, SimCLR and compositing the final-probe numbers differ from these means by less than one point.

Q: Why did F2 work on Spur-CIFAR10 but not MetaShift?

A: 45,000 images cannot be memorized (shuffled targets reach 0.02 explained variance), 1,700 can (0.33).

A Arm name decoder

Arm	Meaning
`simclr`	plain SimCLR; `simclr_lt`: LateTVG's SSL hyperparameters
`f2_std`, `f2_w3`, `f2_w10`	F2 trained head, standardized targets, weight 1 / 3 / 10
`f2_white*`, `*_atyp`, `*_balanced`	whitened targets; atypicality weights; balancing weights
`cbc`, `cbc_plain`	F3 concept blocks; with uniform positive weights
`xfit`, `xfit_norm`	cross-fit; `_norm` = repaired (unit rows, absolute ridge)
`xfit_meaning_w10_holdout`	meaning factors from 5%, weight 10, 20% hold-out
`*_clippca`, `*_response`, `*_residual`	targets: CLIP principal components; dense image responses; residuals
`spatial_w5_holdout`	**SCD** (spatial weight 5, 20% hold-out); `spatial_w5`: without hold-out
`spatial_pca_w5_holdout`	PCA-map control; `_noanimals`, `_noscenes`: concept ablations
`region_w1_holdout`	SCD + region contrast weight 1
`comp_w5_holdout`	SCD + compositing; `comp_random`, `comp_context`: silhouette and context controls
`mix_w5_holdout`	symmetric mixing; `mix_box`: rectangles; `mix_hard`: hard positive; `mix_noregion`: no region term
`heads_w5_holdout`	per-group heads; `heads_single`: one head; `heads_random`: random head
`*_shuffled*`	same loss on another image's targets or maps

B Where things live

What	Path
Research log (dated, every round)	`docs/notes/CONCEPT_FACTORS_2026_09_26.md`
Phase A history (LaTeX)	`docs/xfit/xfit_history.tex`
SCD paper draft	`paper/SpatialConcepts.tex`
Method code (F2, xfit, SCD, region, compositing, mixing, heads)	`cospro/methods/concept_factors.py`
Concept maps (ClearCLIP)	`cospro/pipeline/concept_maps.py`, `scripts/build_concept_maps.sbatch`
Meaning grouping	`cospro/cli/build_meaning_groups.py`
Object/scene text scores	`cospro/pipeline/concept_type.py`
Study launcher (arms, at most 8 jobs)	`scripts/submit_factor_study.sh`
Result tables per dataset	`outputs/reports/factors_<dataset>/summary.md`
Figures of this report	`docs/meeting_2026_11_10/make_figures.py`

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